Chapter 1
Conditional Probability and Expectation

- Poisson, Exponential and Gaussian random variables
- Conditional Probability and Expectation
- Properties of Conditional Expectation
- Computing Probabilities and Expectations by Conditioning

Review of Probability

- We first review some special random variables which are useful and basic building blocks when constructing and computing stochastic processes.

- Chapter 1 and 2 of Ross give a very comprehensive review of basic probability theory.

Poisson

- A random variable X, taking on one of the values $0, 1, 2, ...$, is said to be a Poisson random variable with parameter λ, if for some $\lambda > 0$

$$P(X = k) = e^{-\lambda}\frac{\lambda^k}{k!}, \text{for } k \geq 0.$$

- If $n >> 0$, $np = \lambda$, then a Poisson(λ) can be used to approximate a Binomial(n, p).

- $E[X] = \lambda$, Var$[X] = \lambda$.

- **Additive Property:** If X and Y are two independent Poisson r.v.'s with parameters λ_1 and λ_2 respectively, Then $Z = X + Y$ is a Poisson r.v. with parameter $\lambda_1 + \lambda_2$.

$X_i \sim Poi(\lambda_i)$, $i = 1, 2, \cdots, n$ $\Rightarrow$ $X_1 + X_2 + \cdots + X_n \sim Poi(\lambda_1 + \lambda_2 + \cdots + \lambda_n)$

Additive property of Binomial dist :

$X_1 \sim Bin(n_1, p)$
$X_2 \sim Bin(n_2, p)$
$\vdots$
$X_m \sim Bin(n_m, p)$
(They are indept.)

$\Rightarrow X_1 + X_2 + \cdots + X_n \sim Bin(n_1 + n_2 + \cdots + n_m, p)$

Geometric dist

Flip a coin repeatedly until one get the 1st Heads.

$\begin{cases} p = P(\text{Heads}) \\ q = P(\text{Tails}) \end{cases}$

X = the total # of flips, $X = 1, 2, 3 \cdots$

$P(X=k) = P(\underbrace{T \cdots T}_{k-1} H) = p q^{k-1}$, $k = 1, 2, 3, \cdots$

$EX = \frac{1}{p}$ = reciprocal of the success rate. $\quad Var(X) = \frac{q^2}{p}$

eg1. Roll a fair die repeatedly until one gets the 1st Ace (1-dot) $\quad X$ = # of rolls

In this case, $p = \frac{1}{6}$, $q = P(\text{not 1-dot}) = \frac{5}{6}$ $\quad EX = \frac{1}{\frac{1}{6}} = 6$

eg2. A newly married couple like girls.

Family planning: they want k girls then no more kids.

$\begin{cases} P(\text{Girl}) = p \\ P(\text{Boy}) = q \end{cases}$

X = # of kids in the family $\quad X = k, k+1, k+2, \cdots$

$P(X=i) = P($  $) = \binom{i-1}{k-1} p^{k-1} q^{i-k} \cdot p = \binom{i-1}{k-1} p^k q^{1-k}$

$i-1$: $k-1$ girls

referred to

Negative Binomial dist

For the negative binomial dist with parameter k and p,

we can decompose $X = X_1 + \cdots + X_m$

where $X_1, \cdots, X_m$ are indept and each follows a Geometric dist $Geo(p)$.

Therefore, $EX = EX_1 + \cdots + EX_m = \frac{1}{p} + \cdots + \frac{1}{p} = \frac{m}{p}$

Another way: $EX = \sum_{i=m}^{\infty} i P(X=i) = \sum_{i=m}^{\infty} i \binom{i-1}{k-1} p^k q^{n-k} = \cdots = \frac{m}{p}$



Exponential

- A random variable X, taking value on $[0, \infty)$ is said to be an exponential random variable with *rate* λ, if its probability density function equals

$$f_X(x) = \lambda e^{-\lambda x}.$$

Then, its tail

$$P(X \geq x) = e^{-\lambda x}.$$

pdf: $f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{o.w.} \end{cases}$

- $E[X] = \frac{1}{\lambda}$, $\text{Var}[X] = \frac{1}{\lambda^2}$

- **Memoryless Property:** For any $a > 0$,

$$P(X > a + x | X > a) = e^{-\lambda x}$$

$P(X - a > b \mid X > a) = P(X > b)$

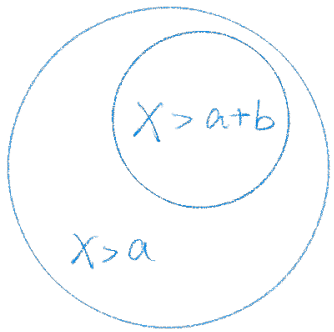
In other words, $X - a$ condition on that $X > a$ is still an exponential r.v. with rate λ.

why?

$$P(X > x) = \int_x^\infty \lambda e^{-\lambda t}\, dt = -e^{-\lambda t}\Big|_x^\infty = e^{-\lambda x}$$

$$\text{Then } P(X - a > b \mid X > a) = \frac{P(X - a > b,\ X > a)}{P(X > a)} = \frac{P(X > a + b,\ X > a)}{P(X > a)} = \frac{P(X > a + b)}{P(X > a)}$$

$$= \frac{e^{-\lambda(a+b)}}{e^{-\lambda a}}$$



Remark:

- In discrete family, the Geometric dist has the memoryless property:

  Let $X \sim Geo(p)$

  $$P(X-a>b \mid X>a) = \frac{P(X>a+b)}{P(X>a)} = \frac{q^{a+b}}{q^{a}} = q^{b} = P(X>b)$$

  $$P(X>a) = P(\overbrace{BB\cdots B}^{a}) = q^{a}$$

  In other words, $P(X-a>b \mid X>a) = P(X>b)$

  In this case, $X = 1, 2, 3\cdots$ (X is discrete)

- Exponential dist. is the member in the cont. family which has the memoryless property.

Gaussian

- A random variable X, taking value on $(-\infty, \infty)$ is said to be an Gaussian/normal random variable with parameter (μ, σ^2), if its probability density function equals

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

- $E[X] = \mu$, $\mathrm{Var}[X] = \sigma^2$.

- **Linearity Property:** If $X \sim N(\mu, \sigma^2)$, for any $a, b \in \mathbb{R}$,

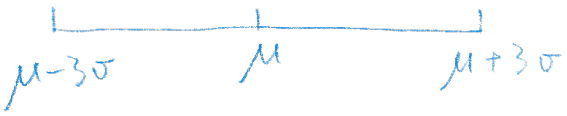
$$aX + b \sim N(a\mu + b, a^2\sigma^2).$$

This property implies
if $X \sim N(\mu, \sigma^2)$ then
$\frac{X-\mu}{\sigma} \sim N(0,1)$

$P(-\sigma < X-\mu < \sigma) = 68\%$

$P(-2\sigma < X-\mu < 2\sigma) = 95\%$

$P(\mu-3\sigma < X < \mu+3\sigma) = 99.75\%$



Conditional Expectation

- One of the most useful concepts in probability theory is that of conditional probability and conditional expectation, especially in the study of stochastic process (also in time series analysis).

- Conditional expectation describe the dependence of multiple random variables. For example, the future conditional distribution of a stochastic process condition on the past observations.

- The mathematical definition of condition expectation reflects the idea that people's estimation on a random outcome changes with the information they obtain.

Discrete case

- If X and Y are discrete random variables, the conditional expectation of a function of X, $g(X)$ given that $Y = y$, is

$$E[g(X)|Y = y] = \frac{E[g(X)1(Y = y)]}{P(Y = y)} = \frac{\sum_{x \in \mathcal{S}_X} g(x)P(X = x, Y = y)}{\sum_{x \in \mathcal{S}_X} P(X = x, Y = y)}.$$

- If $g(x) = 1(x = x_0)$, the conditional expectation $E[f(X)|Y = y]$ becomes the conditional probability that $X = x_0$ given $Y = y$:

$$P(X = x_0|Y = y) = E[1(X = x_0)|Y = y].$$

Example

- **Example 3.3:** If X and Y are independent Poisson random variables with respective means λ_1 and λ_2, calculate the conditional expected value of X given that $X + Y = n$.

eg1. 1st we compute the conditional pmf of X given $X+Y$ $\boxed{P(X=k \mid X+Y=n)}$

We know $X \sim Poi(\lambda_1)$, $Y \sim Poi(\lambda_2)$

and X & Y are indept

By def, $P(X=k \mid X+Y=n) = \dfrac{P(X=k, X+Y=n)}{P(X+Y=n)}$

we know $X+Y \sim Poi(\lambda_1+\lambda_2)$

The above is equal to $\dfrac{P(X=k, Y=n-k)}{P(Poi(\lambda_1+\lambda_2)=n)}$

indep. $\dfrac{P(X=k)\cdot P(Y=n-k)}{P(Poi(\lambda_1+\lambda_2)=n)} = \dfrac{\frac{\lambda_1^k}{k!}e^{-\lambda_1}\,\frac{\lambda_2^{n-k}}{(n-k)!}e^{-\lambda_2}}{\frac{(\lambda_1+\lambda_2)^n}{n!}e^{-(\lambda_1+\lambda_2)}}$

Significantly, we get $\dfrac{n!}{k!(n-k)!}\left(\dfrac{\lambda_1}{\lambda_1+\lambda_2}\right)^k\left(\dfrac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k}$

$\binom{n}{k}$

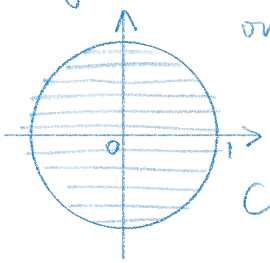
In summary, given $X+Y=n$, $X \sim Bin\left(n, \dfrac{\lambda_1}{\lambda_1+\lambda_2}\right)$

The conditional mean of X given $X+Y=n$ is $n\dfrac{\lambda_1}{\lambda_1+\lambda_2}$

Conditional variance $Var(X \mid X+Y=n) = n\cdot\dfrac{\lambda_1}{\lambda_1+\lambda_2}\cdot\dfrac{\lambda_2}{\lambda_1+\lambda_2}$

Sometimes we write $\begin{cases} E(X \mid X+Y) = \dfrac{\lambda_1}{\lambda_1+\lambda_2}(X+Y) \\ Var(X \mid X+Y) = \dfrac{\lambda_1\lambda_2}{(\lambda_1+\lambda_2)^2}(X+Y) \end{cases}$

eg2. Dart game Let (X,Y) have uniform dist over $\{(x,y); x^2+y^2 \le 1\}$ i.e., the pdf $x^2 \le 1-y^2$



$f(x,y) = \dfrac{1}{\text{area of the shape}} = \dfrac{1}{\pi}$, $x^2+y^2 \le 1$

Compute the conditional pdf of X given Y

$\boxed{f_{X|Y}(x|y) = \dfrac{f(x,y)}{f_Y(y)}}$

Wigner's Semi-Circle Law

Now $f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \frac{1}{\pi}\,dx = \frac{2}{\pi}\sqrt{1-y^2}$

Then the conditional pdf of X given Y is

$f_{X|Y}(x|y) = \dfrac{\frac{1}{\pi}}{\frac{2}{\pi}\sqrt{1-y^2}} = \dfrac{1}{2\sqrt{1-y^2}}$, $-\sqrt{1-y^2} \le x \le \sqrt{1-y^2}$

variable! given!

Thus X given $Y=y$ follows the uniform dist over $(-\sqrt{1-y^2}, \sqrt{1-y^2})$

$\begin{cases} E(X \mid Y=y) = 0 \quad \text{or} \quad E(X \mid Y) = 0 \\ Var(X \mid Y=y) = \dfrac{(2\sqrt{1-y^2})^2}{12} = \dfrac{1-y^2}{3} \quad \text{or} \quad Var(X \mid Y) = \frac{1}{3}(1-Y^2) \end{cases}$

eg3. (X,Y) has pdf $f(x,y) = \begin{cases} x+y, & 0\le x,y\le 1 \\ 0, & \text{o.w.} \end{cases}$

Compute $E(X|Y)$

Solution: We need to compute the conditional pdf $f(x|y)$

1st, Y has pdf $f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$

$= \int_0^1 (x+y)\,dx = y+\frac{1}{2}$

$\therefore f_Y(y) = \begin{cases} y+\frac{1}{2}, & 0\le y\le 1 \\ 0, & \text{o.w.} \end{cases}$

Then $f(x|y) = \dfrac{f(x,y)}{f_Y(y)} = \dfrac{x+y}{\frac{1}{2}+y}$, $0\le x\le 1$

eg. $\begin{cases} f(x|y=\frac{1}{2}) = x+\frac{1}{2}, & 0\le x\le 1 \\ f(x|y=0) = 2x, & 0\le x\le 1 \\ f(x|y=-2) = 0 \end{cases}$

Given $y=\frac{1}{2}$, the pdf of X is

$x+\frac{1}{2}$, $x\in[0,1]$

The mean is $\int_0^1 x(x+\frac{1}{2})\,dx = \frac{7}{12}$

Given $Y=y$, the conditional mean of X:

$\int_0^1 \frac{x+y}{\frac{1}{2}+y}\,dx = \frac{1}{y+\frac{1}{2}} \int_0^1 (x^2+xy)\,dx = \frac{3y+2}{6y+3}$

$\therefore E(X|Y) = \dfrac{3Y+2}{6Y+3}$

Conditional Expectation as Random Variable

- The value of $E[X|Y=y]$ depends only on y. Therefore, the conditional expectation can be viewed as a function of the random variable Y if we do not specify y. In this case, we write it as $E[g(X)|Y]$ and it is a random variable by itself.

- Let X be the output of throwing a dice and $Y = 1(X \text{ is odd})$. Then we know that $P(Y=0) = P(Y=1) = 0.5$. We can easily compute that

$$E[X|Y=0] = 4, \text{ and } E[X|Y=1] = 3.$$

  Therefore, if we define $Z = E[X|Y]$, we can write

$$Z = 3 + 1(Y=0),$$

  and hence $P(Z=3) = P(Z=4) = 0.5$.

Continuous case

- If X and Y are continuous random variables with joint probability density function $f(x, y)$, the *conditional expectation* of a function of X, $g(X)$ given that $Y = y$, is

$$E[g(X)|Y = y] = \frac{E[g(X)1(Y = y)]}{P(Y = y)} = \frac{\int_{-\infty}^{\infty} g(x)f(x, y)dx}{\int_{-\infty}^{\infty} f(x, y)dx}.$$

  The conditional density function is given as

$$f_{X|Y}(x|y) = \frac{f(x, y)}{\int_{-\infty}^{\infty} f(x, y)dx}.$$

  Moreover,

$$E[g(X)|Y = y] = \int_{-\infty}^{\infty} g(x)f_{X|Y}(x|y)dx.$$

- Similarly, the conditional expectation $E[g(X)|Y]$ can be viewed as a function of Y and thus is a random variable by itself.

Example

- **Example 3.8:** Let X_1 and X_2 be independent exponential random variables with rates μ_1 and μ_2. Find the conditional density of X_1 given that $X_1 + X_2 = t$.

joint pdf of X_1 and X_1+X_2

$$f_{X_1, X_1+X_2}(x_1|t) = \frac{f(x_1, t)}{f_{X_1+X_2}(t)}$$

X_1 has pdf $f_1(x_1) = \mu_1 e^{-\mu_1 x_1}$

X_2 has pdf $f_2(x_2) = \mu_2 e^{-\mu_2 x_2}$

The joint pdf of X_1 & X_2:

$$f(x_1, x_2) = \begin{cases} \mu_1\mu_2 e^{-\mu_1 x_1 - \mu_2 x_2}, & x_1 \geq 0, x_2 \geq 0 \\ 0, & \text{o.w.} \end{cases}$$

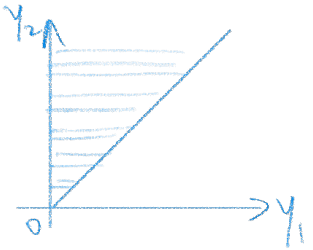
$$\begin{cases} Y_1 = X_1 \\ Y_2 = X_1 + X_2 \end{cases} \quad 0 \leq Y_1 \leq Y_2$$

Solve X_1, X_2 $\begin{cases} X_1 = Y_1 \\ X_2 = Y_2 - Y_1 \end{cases}$

$$J = \det\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} = 1$$

The joint pdf of Y_1 & Y_2 is

$$g(y_1, y_2) = \mu_1\mu_2 e^{-\mu_1 y_1 - \mu_2(y_2 - y_1)}, \quad 0 \leq y_1 \leq y_2$$



Y_2 has pdf $g_{Y_2}(y_2) = \int_{-\infty}^{\infty} g(y_1, y_2)\, dy_1$

$$= \mu_1\mu_2 \int_0^{y_2} e^{-\mu_1 y_1 - \mu_2(y_2 - y_1)}\, dy_1 = \frac{\mu_1\mu_2}{\mu_1 - \mu_2}\left[e^{-\mu_2 y_2} - e^{-\mu_1 y_2}\right]$$

Then the conditional pdf of Y_1 given Y_2 is

$$g(y_1|y_2) = \frac{\mu_1\mu_2 e^{-\mu_1 y_1 - \mu_2(y_2 - y_1)}}{\frac{\mu_1\mu_2}{\mu_1 - \mu_2}\left[e^{-\mu_2 y_2} - e^{-\mu_1 y_2}\right]} = \frac{(\mu_2 - \mu_1) e^{-(\mu_1 - \mu_2) y_1}}{1 - e^{-(\mu_1 - \mu_2) y_2}}$$

$$E(Y_1|Y_2) = \frac{\mu_2 - \mu_1}{1 - e^{-(\mu_1 - \mu_2) y_2}} \int_0^{y_2} y_1 e^{-(\mu_1 - \mu_2) y_1}\, dy_1$$

$$= \frac{1}{1 - e^{-(\mu_1 - \mu_2) y_2}} \frac{1}{\mu_1 - \mu_2} \int_0^{(\mu_1 - \mu_2) y_2} x e^{-x}\, dx$$

$$= \cdots$$

Example

(4)

- **Example 3.8:** Let X_1 and X_2 be independent exponential random variables with rates μ_1 and μ_2. Find the conditional density of X_1 given that $X_1 + X_2 = t$.

Solution. Given $X_1 + X_2 = t$, then $0 \le X_1 \le t$.

Set $Y_1 = X_1$, $Y_2 = X_1 + X_2$ $\Rightarrow$ $X_1 = Y_1$, $X_2 = Y_2 - Y_1$, $Y_1 < Y_2$.

Jacobian: $J = \begin{vmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{vmatrix} = \begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix} = 1.$

Thus the joint pdf of Y_1, Y_2 is

$$f_{Y_1,Y_2}(y_1, y_2) = \mu_1 \mu_2 e^{-\mu_1 x_1 - \mu_2 x_2} \Big|_{\substack{x_1 = y_1 \\ x_2 = y_2 - y_1}} \cdot |J| = \mu_1 \mu_2 e^{-\mu_1 y_1 - \mu_2 (y_2 - y_1)}$$

$$= \mu_1 \mu_2 e^{-(\mu_1 + \mu_2) y_1} \cdot e^{-\mu_2 y_2}$$

In summary

$$f(y_1, y_2) = \begin{cases} \mu_1\mu_2 e^{-(\mu_1-\mu_2)y_1} \cdot e^{-\mu_2 y_2}, & 0 \le y_1 \le y_2 \\ 0 & \text{o.w.} \end{cases}$$

Thus, Y_2 has pdf

$$f_{Y_2}(y_2) = \int_{-\infty}^{\infty} f(y_1, y_2)\, dy_1 = \mu_1\mu_2 e^{-\mu_2 y_2} \int_0^{y_2} e^{-(\mu_1-\mu_2)y_1}\, dy_1$$

$$= \mu_1\mu_2 e^{-\mu_2 y_2} \frac{1}{\mu_1-\mu_2}\left[1 - e^{-(\mu_1-\mu_2)y_2}\right]$$

Thus, the conditional pdf $f_{Y_1|Y_2}(y_1|y_2)$

$$= \frac{\mu_1\mu_2 e^{-(\mu_1-\mu_2)y_1} e^{-\mu_2 y_2}}{\mu_1\mu_2 e^{-\mu_2 y_2}\left(1 - e^{-(\mu_1-\mu_2)y_2}\right)} (\mu_1 - \mu_2)$$

$$= \frac{(\mu_1-\mu_2) e^{-(\mu_1-\mu_2)y_1}}{1 - e^{-(\mu_1-\mu_2)y_2}}, \quad 0 \le y_1 < y_2$$

Recall $Y_1 = X_1$, $Y_2 = X_1 + X_2$

Thus, the conditional pdf of X_1 given $X_1 + X_2 = t$ is

(*) $$f(x_1 | x_1 + x_2 = t) = \frac{(\mu_1-\mu_2) e^{-(\mu_1-\mu_2)x_1}}{1 - e^{-(\mu_1-\mu_2)t}}, \quad 0 \le x_1 \le t$$

as $\mu_1 \neq \mu_2$. Then

$$E(X_1 | X_1 + X_2 = t) = \frac{\mu_1 - \mu_2}{1 - e^{-(\mu_1-\mu_2)t}} \int_0^t x_1 e^{-(\mu_1-\mu_2)x_1}\, dx_1$$

Set $y = (\mu_2 - \mu)x_1$. Then

$$(\mu_1-\mu_2)\int_0^t x_1 e^{-(\mu_1-\mu_2)x_1}\,dx_1$$

$$= \frac{-1}{\mu_2-\mu_1}\int_0^{(\mu_2-\mu_1)t} y e^y\,dy$$

$$= \frac{-1}{\mu_2-\mu_1}\left(ye^y - e^y\right)\Big|_0^{(\mu_2-\mu_1)t}$$

$$= \frac{-1}{\mu_2-\mu_1}\left[(\mu_2-\mu_1)te^{(\mu_2-\mu_1)t} - e^{(\mu_2-\mu_1)t} + 1\right]$$

Thus

$$E(X_1 \mid X_1+X_2=t) = \frac{t(\mu_2-\mu_1)e^{(\mu_2-\mu_1)t} - e^{(\mu_2-\mu_1)t} + 1}{\left[1-e^{(\mu_2-\mu_1)t}\right](\mu_2-\mu_1)}$$

$$= \frac{(\mu_2-\mu_1)te^{(\mu_2-\mu_1)t} - e^{(\mu_2-\mu_1)t} + 1}{\left[1-e^{(\mu_2-\mu_1)t}\right](\mu_1-\mu_2)}.$$

If $\mu_1=\mu_2$, check the calculation before, or use (*),

$$f(x_1 \mid x_1+x_2=t) = \frac{1}{t}, \quad 0 \le x_1 \le t.$$

i.e., $x_1 \mid x_1+x_2=t \sim$ Uniform on $[0, t]$.

In this case $E(x_1 \mid x_1+x_2=t) = \frac{t}{2}$.

(*) $f(x_1 \mid x_1+x_2=t) = \frac{(\mu_1-\mu_2)e^{-(\mu_1-\mu_2)x_1}}{1-e^{-(\mu_1-\mu_2)t}}, \quad 0 \le x_1 \le t$

Properties of conditional expectation

- Suppose X and Y are two random variables (in the same probability space). Then we have:
  1. For any constant $a, b \in \mathbb{R}$, $E[aX + b|Y] = aE[X|Y] + b$. (Linearity)
  2. $E[X] = E[E[X|Y]]$. (Mean of conditional mean is the true mean)
  3. If X and Y are independent, $E[X|Y] = E[X]$.

2025. 2. 10 Mon

* Conditional expectation
* Regular Conditional expectation
* Wald identity
* Thinning theorem

Properties of conditional expectation

- The equation $E[X] = E[E[X|Y]]$ is also called "tower" or "telescoping" property of conditional expectation.

- **A trivial example:** Let X be the output of throwing a dice and $Y = 1(X \text{ is odd})$. We have defined $Z = E[X|Y]$.

$Y \sim \text{Ber}(\frac{1}{2})$ ← { $Y=0$ if X is even

$$E[E[X|Y]] = E[Z] = 0.5 \times 3 + 0.5 \times 4 = 3.5.$$

On the other hand,

$$E[X] = \frac{1+2+\ldots+6}{6} = \frac{21}{6} = 3.5.$$

$E[X|Y=1] = \frac{1}{3}\cdot 1 + \frac{1}{3}\cdot 3 + \frac{1}{3}\cdot 5 = 3$

Similarly, $E[X|Y=0] = \frac{1}{3}\cdot 2 + \frac{1}{3}\cdot 4 + \frac{1}{3}\cdot 6 = 4$

Two examples

- **Example 3.10:** Suppose that the expected number of accidents per week at an industrial plant is four. Suppose also that the numbers of workers injured in each accident are independent random variables with a common mean of 2. Assume also that the number of workers injured in each accident is independent of the number of accidents that occur. What is the expected number of injuries during a week?

Two examples

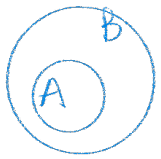
Example 3.12: A miner is trapped in a mine containing three doors. The first door leads to a tunnel that takes him to safety after two hours of travel. The second door leads to a tunnel that returns him to the mine after three hours of travel. The third door leads to a tunnel that returns him to his mine after five hours. Assuming that the miner is at all times equally likely to choose any one of the doors, what is the expected length of time until the miner reaches safety?

Poisson Thinning

- **Example 3.24:** Suppose that the number of people who visit a yoga studio each day is a Poisson random variable with mean λ. Suppose further that each person who visits is, independently, female with probability p or male with probability $1-p$. Find the joint probability that exactly n women and m men visit the academy today.

2025.2.12 Wed

* Last part of Conditional expectation
* Def of Markov chains
  (discrete time, discrete state)



If $A \subset B$: $AB = A$

If $w \in A$, then $w \in B$

X = # of women visits the studio

Y = ··· men ——

$X+Y \sim Poi(\lambda)$

$X \mid X+Y = Bin(X+Y, p)$

$P(X=n, Y=m) = ?$

$P(X=n, Y=m \mid X+Y=m+n)$

$=$

$P(X=n \mid X+Y=m+n)$

$=$

$P(Bin(m+n, p)=n)$

1st, $P(X=n, Y=m \mid X+Y=m+n) = \binom{m+n}{n} p^n q^m$

$P(X+Y=m+n) = \frac{\lambda^{m+n}}{(m+n)!} e^{-\lambda}$

(相乘)

$P(X=n, Y=m, X+Y=m+n) = P(X=n, Y=m) = \cdots = \frac{p^n q^m \lambda^{m+n}}{m!\, n!} e^{-\lambda} = \frac{(\lambda p)^n}{n!} e^{-\lambda p} \cdot \frac{(\lambda q)^m}{m!} e^{-\lambda q}$

法1: Marginal pmf: $P(X=n) = \sum_{m=0}^{\infty} \frac{p^n q^m \lambda^{m+n}}{m!\, n!} e^{-\lambda} = \frac{p^n q^n e^{-\lambda}}{n!} \sum_{m=0}^{\infty} \frac{(\lambda q)^m}{m!}$

$= \frac{(\lambda p)^n}{n!} e^{-\lambda p}$

法2: $P(X=n, Y=m) = P(Poi(\lambda p)=n, Poi(\lambda q)=m) \quad \forall m \geq 0, n \geq 0$

Conclude: $X \sim Poi(\lambda p)$

$Y \sim Poi(\lambda q)$

X & Y are indept

Poisson Thinning

- The generalization of Example 3.23 is called the Poisson thinning theorem, which is very useful for analyzing continuous-time Markov chain.

- **(Poisson Thinning Theorem)** Poisson distributed number of events, N, with mean λ is independently classified as being one of k types, with the probability that it is type i being p_i, $i = 1, ..., k$, $\sum_{i=1}^{k} p_i = 1$. If N_i is the number that are classified as type i, then $N_1, ..., N_k$ are independent Poisson random variables with respective means $\lambda p_1, ..., \lambda p_k$.

Conditional Variance

- The conditional variance of X given $Y = y$, $\text{Var}(X|Y = y)$, which is defined as the variance of the conditional distribution of X given $Y = y$:

$$\text{Var}(X|Y = y) = E[X^2|Y = y] - (E[X|Y = y])^2.$$

- **(The Conditional Variance Formula)**

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}(E[X|Y]).$$

Variance = mean of conditional variance + variance of conditional mean

Eg. Given $X > 0$, $Y \sim$ Uniform dist over $[0, X]$

$E(Y|X) = \frac{X}{2}$

$\text{Var}(Y|X) = \frac{X^2}{12} \neq \text{Var}(Y)$ in general

Example

- **Example 3.20: (Variance of Compound Random Variable)** Let X_1, X_2, . . . be i.i.d. random variables with mean μ and variance σ^2, and assume that they are independent of the nonnegative integer valued r.v. N. the r.v. $S = \sum_{i=1}^{N} X_i$ is called a compound random variable. Find its variance.

$S_n = X_1 + X_2 + \cdots + X_n$

$E(S_N|N=n) = E(S_n|N=n)$

1st, $E(S_n) = n \cdot \mu$ $\quad \therefore E(S_N|N) = N\mu$

2nd, $Var(S_n) = n \cdot Var(X_1) = n\sigma^2$ $\quad \therefore Var(S_n|N=n) = n \cdot Var(X_1) = n\sigma^2$ $\quad \therefore Var(S_N|N) = N\sigma^2$

By the formula, $Var(S_N) = E(N\sigma^2) + Var(N\mu) = \sigma^2 E(N) + \mu^2 \cdot Var(N)$